

Information Theory assignment 2

Alireza Mousazadeh

January 3, 2026

1 3.5

$$X_n = X_{n-1} \oplus X_{n-2} \oplus Z_n$$

$$\rightarrow Z_n = X_n \oplus X_{n-1} \oplus X_{n-2}$$

since Z_n is a deterministic function of X_n , every $\{Z_k\}_1^n$ sequences is achievable from $\{X_k\}_1^n$, also $H(Z^n) = H(X^n)$, because if we know X then Z is obvious.

$$H(Z^n) = \sum_{i=1}^n H(Z_i); \quad H(Z_i) = 1 \text{ bit}$$

$$\rightarrow \sum_{i=1}^n H(Z_i) = n$$

then *entropy rate* becomes:

$$\lim_{n \rightarrow \infty} \frac{1}{n} H(Z^n) = 1$$

2 3.6

codes that are not uniquely decodable cant be prefix-free. so options (a), (c), (d) and (e) may or may not be uniquely decodable.

code	cat. seq.		
{1,0,00}	00	00 ?	0, 0 ?
{1,10,00}	100	1, 00 ?	10, 0x ?
{1,10,01}	101	1, 01 ?	10, 1 ?
{0,01}	001	0, 01	0, 0, 1x (invalid code word: 1x)

from above examples only {0,01} is uniquely decodable.

3 3.9

$$\Pr(X_2 = j | X_1 = i) = \begin{cases} \frac{\rho}{1+\delta}; i = 0, j = 1 \\ \frac{\rho+\delta}{1+\delta}; i = 1, j = 1 \end{cases}$$

$$\rightarrow \mathbf{P} = \begin{bmatrix} \frac{1+\delta-\rho}{1+\delta} & \frac{\rho}{1+\delta} \\ \frac{1-\rho}{1+\delta} & \frac{\rho+\delta}{1+\delta} \end{bmatrix} \rightarrow \pi^T \mathbf{P} = \pi^T; \quad (\pi^T = (\pi_0, \pi_1))$$

(a)

$$\pi^T \mathbf{P} - \pi^T I = 0 \rightarrow \pi^T \begin{bmatrix} \rho & \rho \\ 1-\rho & \rho-1 \end{bmatrix} = \mathbf{0} \xrightarrow{\pi_0=1-\pi_1} \pi_1 = \rho, \pi_0 = 1 - \rho$$

(b)

$$H(\{X_k\}) = H(X_2|X_1)|_\pi = \pi_0 H(X_2|X_1=0) + \pi_1 H(X_2|X_1=1) = \dots$$

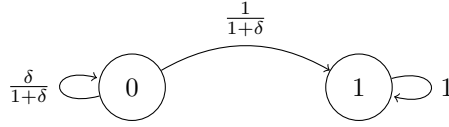
$$(1-\rho) \left(-\frac{\rho}{1+\delta} \log \frac{\rho}{1+\delta} - \frac{1+\delta-\rho}{1+\delta} \log \frac{1+\delta-\rho}{1+\delta} \right) + \rho \left(-\frac{\rho+\delta}{1+\delta} \log \frac{\rho+\delta}{1+\delta} - \frac{1-\rho}{1+\delta} \log \frac{1-\rho}{1+\delta} \right)$$

(c)

$$\rho_m = H(X_1) - H(X_2|X_1) = 1 - 0.81 = 0.19, \quad \rho_t = \rho_m, \quad \rho_d = 0$$

(d) for $\rho = 1$:

$$\mathbf{P} = \begin{bmatrix} \frac{\delta}{1+\delta} & \frac{1}{1+\delta} \\ 0 & 1 \end{bmatrix}$$



if we start from state $X_n = 0$, it will transit to state $X_{n+1} = 1$ and never returns. so process isn't **irreducible**. and $H(\{X_k\}) = 0$.

(e) for $\delta = 0$ *probability transition matrix* rank become 1 and $X_2 \perp X_1$, and entropy becomes $H(X) = -\rho \log(\rho) - (1-\rho) \log(1-\rho)$

(f) computing *transition matrix* for $\rho = 0.5, \delta = 1.5$:

$$\rightarrow \Pr(1|0) = 0.2, \Pr(1|1) = 0.8; \pi = (0.5, 0.5)^T$$

$$\rightarrow H(\mathcal{X}) = 0.5h_b(0.2) + 0.5h_b(0.8) \approx 0.72$$

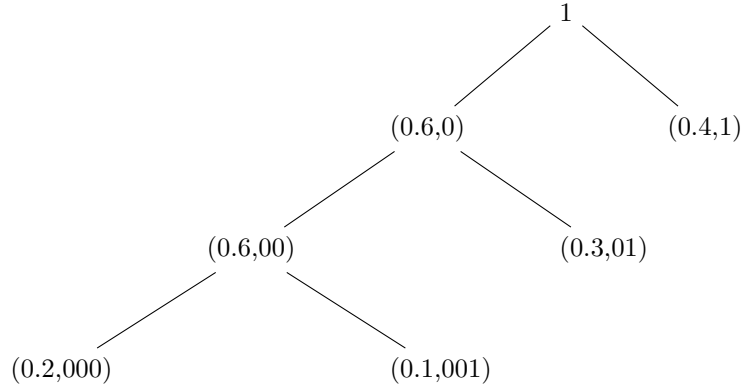
- First order: $\mathcal{X} \in \{0, 1\}, P(0) = P(1) = 0.5, R_1 = 1$
- Second order: $\mathcal{X} \in \{00, 01, 10, 11\}, R_2 = 0.75$
- Third order: $\mathcal{X} \in \{000, 001, 010, 011, \dots\}, R_3 = 0.73$

$$R_1 > R_2 > R_3 > \dots > 0.72 = H(\mathcal{X})$$

4 3.11

(a) starting from lowest probability we have:

code	Z	Probability			
1	e1	0.4	0.4	0.6	1
01	e2	0.3	0.3	0.4	
001	e3	0.2	0.3		
000	e4	0.1			



$$E(L) = 1.9 \text{ bits per symbol}$$

(b)

$$\begin{cases} \Pr(U_2 = 0|U_1 = 1) \rightarrow Z_1 = e_1 \\ \Pr(U_2 = 0|U_1 = 0) \rightarrow Z_1 \in \{e_2, e_3, e_4\} \end{cases} \longrightarrow \Pr(U_2 = 0|U_1 = 1) \neq \Pr(U_2 = 0|U_1 = 0)$$

(c)

$$E(l) = 0.4 + (0.3)2 + (0.2)3 + (0.1)3 = 1.9 \geq 1.846 = H(Z)$$

$$E(l) \geq H(Z) \iff H(Z) \leq E(l) \leq H(Z) + 1 \quad \diamond$$

equality holds when probabilities are powers of 2.

(d)

5 3.13

(a)

if $P(X = x_i) = 2^{-n_i}$ then $\sum P(X = x_i) = \sum 2^{-n_i} = 1$, so by Kraft inequality there is a prefix code with length n_i :

$$H(X) = -\sum_i 2^{-n_i} \log_2(2^{-n_i}) = -\sum_i 2^{-n_i}(-n_i) = \bar{L}$$

from problem 12: $\frac{1}{n} \sum_i 2^{-n_i} n_i \leq H(X) + \frac{1}{n}$ and by TH. 3.22 $\frac{1}{n} H(X) \leq \frac{1}{n} \sum_i 2^{-n_i} n_i \leq H(X) + \frac{1}{n}$:

$$\rightarrow H(X) \leq \sum_i 2^{-n_i} n_i \leq H(X) + 1$$

$$H(X) = \bar{L} \iff l_i = \log_2 \left(\frac{1}{P_X(x_i)} \right)$$

l_i is integer so $P_X(x_i) = 2^{-l_i}$ $l_i = n_i \in \mathbf{Z}^+ \diamond$

(b)

for ternary Huffman code same conditions holds in base 3; that is the probability of all symbols must be powers of $1/3$.

(c)

say $\bar{R} = H(X) + 1/2$ then for pair source $H(X, X) + 1 = 2H(X) + 1$ and for equality in right hand $P_{XX}(x, y) = P_X P_X = 2^{-n_{xy}}$ by fixing y , $P_X(x)$ must be powers of $1/2$ and $\bar{R} = H(X) \neq H(X) + 1/2$

6 3.19

(a)

by chain rule:

$$H(X_{n+1}^{2n} | X^n) = \sum_{i=1}^n H(X_{n+i} | X^n, X_{n+1}^{2n-n-1+i}) = \sum H(X_{n+i} | X_1, \dots, X_{n+i-1})$$

for stationary process:

$$\begin{aligned} H(X_{n+i} | X_1, \dots, X_{n+i-1}) &\leq H(X_i | X_1, \dots, X_{i-1}) \\ \rightarrow H(X_{n+1}^{2n} | X^n) &\leq \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1}) = H(X_1, \dots, X_n) \end{aligned}$$

$H(X_1, \dots, X_n)$ is **nonincreasing** so do $H(X_{n+1}^{2n} | X^n)$.

(b)

by previous chain rule expansion we can show that:

$$\begin{aligned} H(X_{n+i} | X_1, \dots, X_{n+i-1}) &\geq H(X_{2n} | X_1, \dots, X_{2n-1}) \\ \rightarrow H(X_{n+1}^{2n} | X^n) &\geq \sum H(X_{2n} | X_1, \dots, X_{2n-1}) = nH(X_{2n} | X_1, \dots, X_{2n-1}) \end{aligned}$$

$$\boxed{nH(X_{2n} | X_1, \dots, X_{2n-1}) \leq H(X_{n+1}^{2n} | X^n) \leq H(X_1, \dots, X_n)} \quad (1)$$

(c)

by tacking limit of [1] as $n \rightarrow \infty$:

$$\boxed{\lim_{n \rightarrow \infty} \frac{1}{n} H(X_{n+1}^{2n} | X^n) = H(\mathcal{X})}$$

$$\frac{1}{n} I(X_{n+1}^{2n}; X^n) = 1/n H(X_{n+1}^{2n}) - 1/n H(X_{n+1}^{2n} | X^n)$$

$$\xrightarrow{n \rightarrow \infty} 1/n H(X_{n+1}^{2n}) - H(\mathcal{X})$$

assuming the process is stationary $H(X_{n+1}^{2n}) = H(X^n)$, then:

$$\boxed{\lim_{n \rightarrow \infty} \frac{1}{n} I(X_{n+1}^{2n}; X^n) = 0}$$

(d)

for stationary Markov process:

$$H(X_{n+1}^{2n}|X^n) = \sum H(X_{n+i}|X_1, \dots, X_{n+i-1}) = \sum H(X_{n+i}|X_{n+i-1}) = \sum H(X_2|X_1)$$

$$\frac{1}{n}H(X_{n+1}^{2n}|X^n) = H(X_2|X_1); \boxed{\lim_{n \rightarrow \infty} H(X_2|X_1) = H(\mathcal{X})}$$